

Next Generation Data Centers and Energy Management Systems

Lecture 5A: Liquid Cooling & Advanced Heat Transfer

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Learning Plan

- Introduction to Data Center Systems and Thermodynamics
- Psychrometrics and Heat Transfer
- Performance Metrics and Standards
- Applied Computational Fluid Dynamics (CFD)
- **Liquid Cooling & Advanced Heat Transfer**
- Chip Cooling: Heat Conduction and Thermal Interface Materials
- Power Delivery and Infrastructure (Energy sources: Geothermal, Solar, Nuclear, etc.)
- Overview of Simulation Field; the Concept of a Digital Twin
- Machine Learning and Optimization Tools
- Modeling and Simulation of an Energy Management System
- Control Systems and Optimization
- Spacecraft Thermal Control
- Final Project Work, Peer Feedback, and Mentor Check-in
- Symposium

Learning Objectives

- Define the Blasius and Polhausen equations, which are used for microelectronics cooling.
- Solve the Blasius and Polhausen equations.

Typical layout of a data center facility

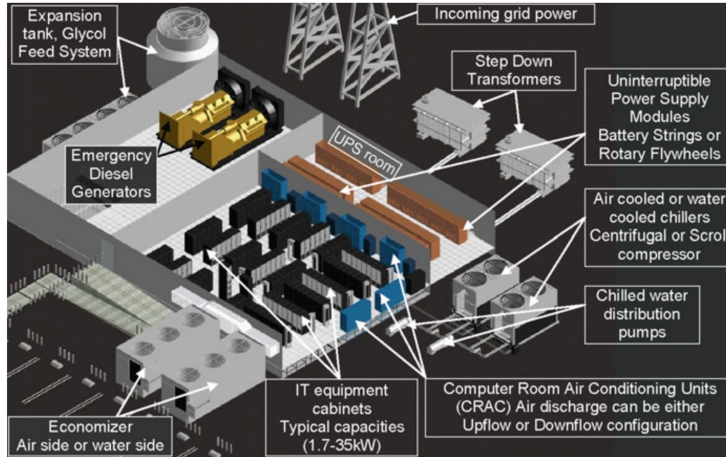


Figure 1: (Figure 1.1 in Ref. [1]).

Seminar 2

- Dr. David Geb, Synopsys (formerly Ansys)
- Title: TBD
- Date: Tuesday, 9/29/26
- Time: 15:30 (Berkeley Time)
- Location: GSPP 150

Project 3

In Project 3, you will analyze a conduction-convection problem. For this you will need to know the basics of CFD and boundary layer theory. You will use ANSYS Fluent.

Projects

We would like to remind you that you are required to follow the report template. Part of the purpose of the projects is for you to have professional examples of your work.

Microelectronics cooling: Solution to Blasius

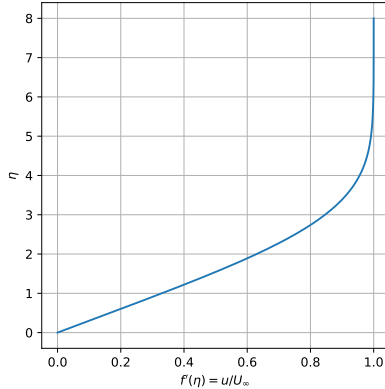


Figure 2: Blasius profile. Solved numerically in 1908. Read his biography [here](#).

Microelectronics cooling: Solution to Pohlhausen

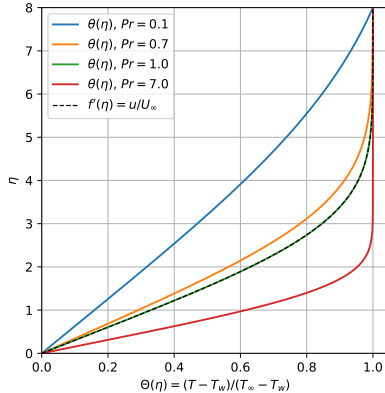


Figure 3: Solution to the Pohlhausen equation. Read his biography [here](#).

References I

- [1] Yogendra Joshi and Pramod Kumar, editors.
Energy Efficient Thermal Management of Data Centers.
Springer US, 2012.